

Longitudinal Study of Solanezumab’s Effect on Memory Decline in Preclinical Alzheimer’s

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Abstract

Preclinical Alzheimer’s disease poses a critical challenge because subtle cognitive decline begins but remains untreated. To address this, we study longitudinal data from a phase 3 double-blind, placebo-controlled trial to determine whether solanezumab can slow memory decline. The data comes from Anti-Amyloid Treatment in Asymptomatic Alzheimer’s (A4) Study. Using the preclinical Alzheimer’s cognitive composite (PACC) score as the outcome variable, we used linear mixed-effects models to identify time trends, covariance structures, and treatment effects. While the PACC scores showed a quadratic decline over time, treatment with solanezumab did not exhibit a statistically significant effect ($p\text{-value} = 0.826$) on slowing this decline.

1 Introduction and Exploratory Data Analysis

A cohort of subjects with preclinical Alzheimer’s was enrolled to test whether solanezumab, a drug targeting amyloid plaques, can slow memory decline and they were expected to be followed for 240 weeks, each occurring at 24-week intervals. The resulting data set includes 1147 subjects and 10822 observations. Key predictors are the time in years since the baseline visit and the treatment they randomly received. Preclinical Alzheimer’s cognitive composite (PACC) composite score is the response variable, where lower scores indicate greater cognitive decline. The PACC score is calculated by summing standardized z-scores derived from four cognitive tests. 73% of subjects completed 11 visits, which results in unbalanced data with the number of PACC measurements varying between 1 and 11, and a median of 5.

We removed time points after 240 weeks with extremely low visit numbers and transformed time from weeks to years. We also rounded time in years to the nearest 0.5-year interval for clear visualization. Figure 1 shows that PACC scores have a slightly decreasing trend over time for two treatment groups, with the solanezumab group exhibiting a faster rate of decline. As shown in Figure 2, the two groups illustrate similar mean and variability patterns throughout the study.

2 Time-Only Longitudinal Analysis

We first identify the most appropriate time trend by utilizing linear mixed-effects models (LMM) and comparing four different time trend models, as summarized in Table 1. The quadratic time trend has the smallest AIC, BIC, and deviance values so we select the quadratic time trend as the best time trend. Furthermore, from Table 2, the random intercept, slope, and quadratic model (RIASQ) shows the lowest AIC, BIC, and deviance values; therefore, we select the RIASQ model with quadratic time trend as the best-fitting model to describe PACC scores' trend over time.

After determining time trends and covariance structure, we obtain the rates at the population level (Table 3). The model indicates that PACC scores change non-linearly over time. The PACC scores show an average increase of 0.285 units per year initially, with the rate of change decreasing by 0.109 units per year, and eventually reversing to a decline. This aligns with the rough quadratic time trend observed in Figure 2. Both p-values are equal to 0, which shows these effects are highly significant and indicates strong evidence for the quadratic temporal trend in PACC scores' decline.

3 Longitudinal Analysis with Time and Treatment

In this section, we include treatment as a predictor and fit the model based on the best-fitting model structure from Section 2, quadratic time trend and RIASQ covari-

ance structure. Table 4 shows that the model with only a quadratic time effect and treatment main effect has the lowest AIC, BIC, and deviance values so this model is identified as the best model. From Table 5, PACC scores decline over time in a pattern consistent with the results in Section 2, with statistically significant linear (p-value = 0) and quadratic (p-value = 0) time effects. This consistency may be because the time trend and covariance structure remain unchanged, and the treatment variable reflects baseline assignment rather than a time-varying effect. Therefore, treatment of solanezumab does not have a significant impact on PACC scores' decline, as indicated by the high p-value (0.826). Randomization of the clinical trial will ensure treatment and control groups are balanced at baseline so that the treatment effect will not be influenced by other factors.

4 Conclusions

We compared two best-fitting models from Section 2 and Section 3, the RIASQ model with a quadratic time effect and the RIASQ model with a quadratic time effect and treatment main effect. Based on the AIC, BIC, and deviance values in Table 6, we conclude that the RIASQ model only with a quadratic time effect has a better fit and is more appropriate for this cohort of subjects. The diagnostic plot further supports this conclusion. The residuals in Figure 3 are centered around the zero line with no special pattern and the variances are constant, indicating that the model fits the data reasonably well. Therefore, PACC scores' decline over time follows a quadratic trend, and the treatment of solanezumab does not show a significant effect in slowing the decline effectively.

Appendix

Table 1: ANOVA table for model comparison between different time trends. We fit four different time trend models with random intercepts. The models include (1) no time trend, (2) linear time trend, (3) quadratic time trend, and (4) cubic time trend.

Model	Parameters	AIC	BIC	Deviance	p-value
Constant	3	46761	46783	46755	–
Linear	4	46455	46484	46447	< .0001
Quadratic	5	46334	46370	46324	< .0001
Cubic	6	46343	46387	46331	0.006

Table 2: ANOVA table for model comparison between different covariance structures. We fit six different covariance models with quadratic time trends. The models include (1) random intercept only, (2) random slope only, (3) Autoregressive AR(1), (4) random intercept and slope, (5) ARMA(1,1), and (6) random intercept, slope, and quadratic.

Model	Parameters	AIC	BIC	Deviance	p-value
model_RI	5	46334	46370	46324	–
model_RS	5	47216	47252	47206	–
model_AR1	6	44515	44558	44503	< .0001
model_RIAS	7	42568	42619	42554	< .0001
model_ARMA	7	43611	43662	43597	–
model_RIASQ	10	42329	42402	42309	< .0001

Table 3: Fixed effects parameter estimates from the RIASQ model for PACC scores’ decline over time. The estimates include the intercept, linear time effect, and quadratic time effect, along with their respective standard errors and test statistics.

Parameter	Coefficient	Standard Error	t-value	p-value
Intercept	-0.211	0.076	-2.8	0.006
Year	0.285	0.035	8.1	0
Year ²	-0.109	0.008	-14.1	0

Table 4: ANOVA table for model comparison between different fixed effect structures. We fit three different RIASQ models to explore the relationship between PACC scores and time, incorporating treatment. The models include (1) a quadratic time effect and treatment main effect, (2) a quadratic time effect, a treatment main effect, and interaction between linear time effect and treatment, and (3) a quadratic time effect, a treatment main effect, and interactions between both linear and quadratic time effect and treatment.

Model	Parameters	AIC	BIC	Deviance	p-value
Year + Year ² + Treatment	11	42333	42413	42311	–
Year*Treatment + Year ²	12	42337	42424	42313	0.139
Year*Treatment + Year ² *Treatment	13	42345	42439	42319	0.015

Table 5: Fixed effects parameter estimates for the RIASQ model with treatment main effect. The estimates include the intercept, linear time effect, quadratic time effect, and treatment main effect, along with their respective standard errors and test statistics.

Parameter	Coefficient	Standard Error	t-value	p-value
(Intercept)	-0.227	0.104	-2.2	0.030
Year	0.285	0.035	8.1	0
Year ²	-0.109	0.008	-14.1	0
Treatment: Solanezumab	0.032	0.145	0.2	0.826

Table 6: ANOVA table for model comparison. The table compares the final models obtained from Section 2 and Section 3: the RIASQ model with quadratic time effect and the RIASQ model with quadratic time effect and treatment main effect.

Model	Parameters	AIC	BIC	Deviance	p-value
Year + Year ²	10	42329	42402	42309	–
Year + Year ² + Treatment	11	42333	42413	42311	0.159

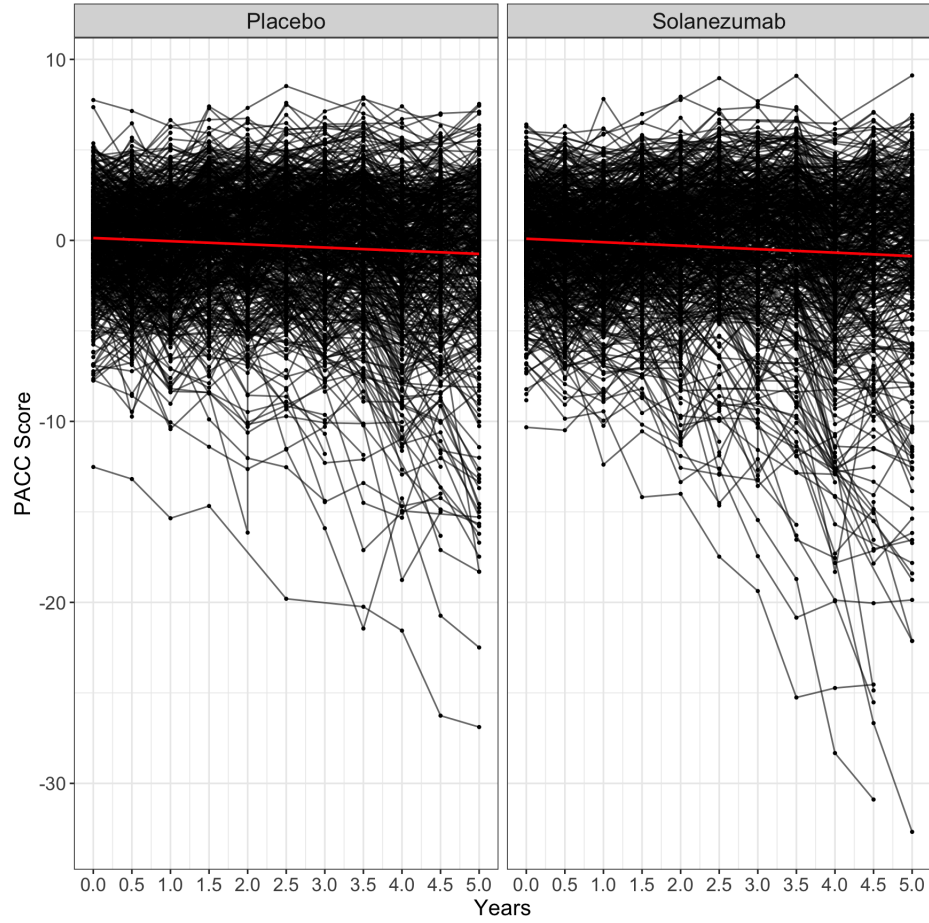


Figure 1: Profile plot of PACC scores. The plot shows individual trajectories of PACC scores over the follow-up period for the two treatment groups: placebo and solanezumab. Measurements are indicated by dot points, and a red line represents the linear trend fitted to all measurements within each group. The x-axis shows the follow-up time in years, and the y-axis shows PACC scores.

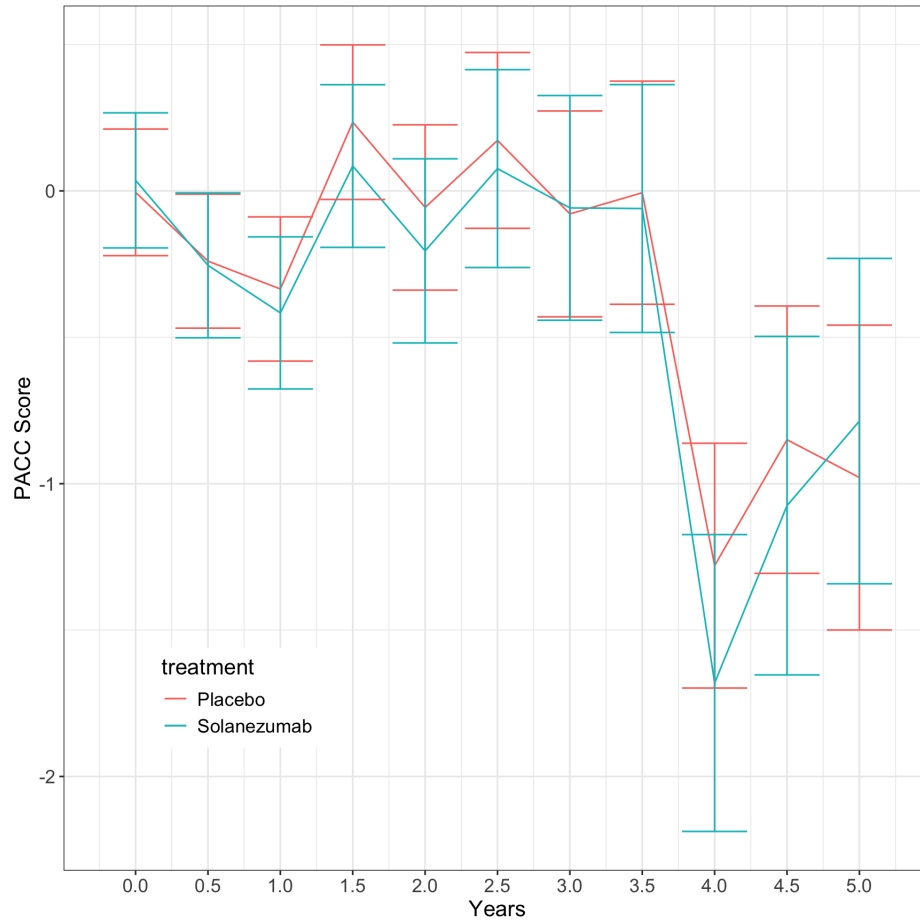


Figure 2: Empirical summary plot of PACC scores. The plot displays the trend of the mean PACC scores over the follow-up time for the two treatment groups, with error bars representing ± 2 standard errors from the mean. The x-axis shows the follow-up time in years, and the y-axis shows the PACC scores.

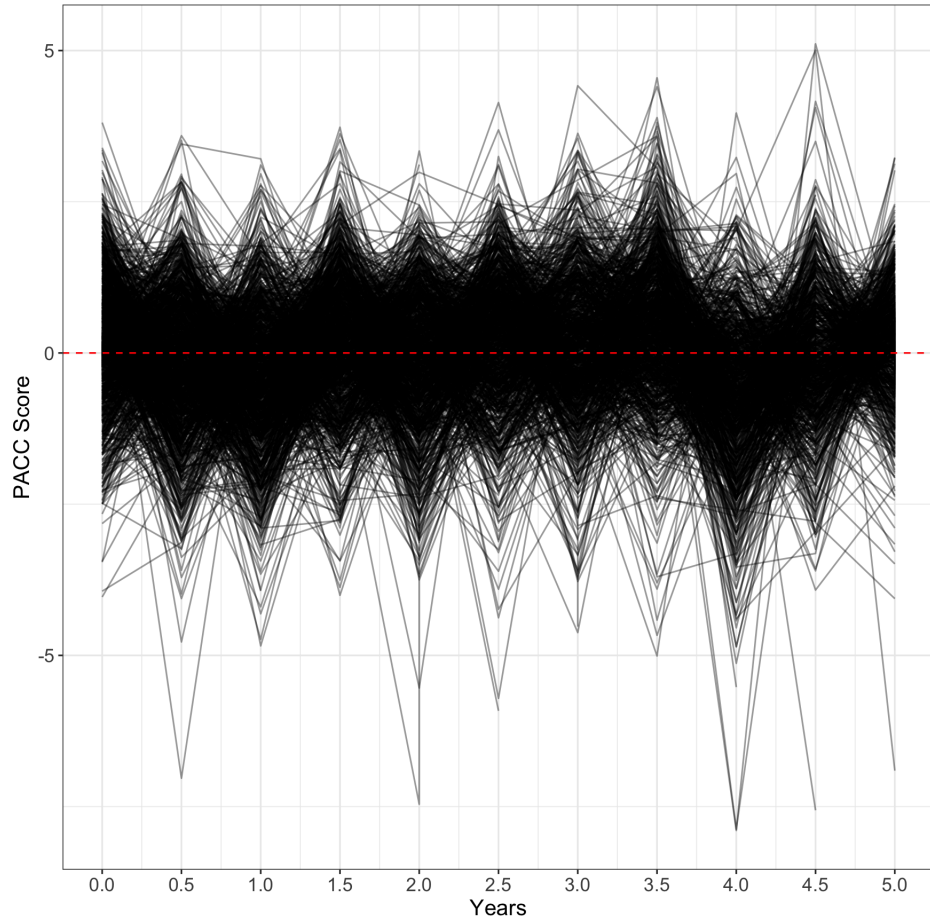


Figure 3: Residual plot for the RIASQ model with a quadratic time trend. This plot shows the model residuals for PACC scores against the time.